Practical No 2-A

Fragments and Activity Interaction

Aim: Develop an Android application to demonstrate Fragment lifecycle methods.

Fragment

A Fragment is a reusable portion of an Activity that represents a part of the user interface and can be combined with other fragments in a single activity.

Fragment Lifecycle Methods

Just like an Activity, a Fragment also has a lifecycle.

onAttach()
↓
onCreate()
↓
onCreateView()
↓
onViewCreated()
↓
onStart()
↓
onResume()

(App Running)

onPause()
↓
onStop()
↓
onDestroyView()
↓
onDestroy()
↓
onDetach()

Code:

1) **MainActivity.java (Keep as it is)**

```
package com.example.nepnewpract;

import android.os.Bundle;
```

```
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       EdgeToEdge.enable(this);
       setContentView(R.layout.activity_main);
       ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v,
insets) -> {
           Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
           v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
           return insets;
       });
   }

}
```

2) MyFragment.java

```
package com.example.twoa;

import android.content.Context;
import android.os.Bundle;

import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;

import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Toast;
```

```
/**
* A simple {@link Fragment} subclass.
* Use the {@link MyFragment#newInstance}
factory method to
* create an instance of this fragment.
*/
public class MyFragment extends Fragment {
    @Override
    public void onAttach(@NonNull Context
context) {
        super.onAttach(context);
        Toast.makeText(getActivity(),
"onAttach()", Toast.LENGTH_SHORT).show();
    }

    // TODO: Rename parameter arguments,
choose names that match
    // the fragment initialization
parameters, e.g. ARG_ITEM_NUMBER
    private static final String ARG_PARAM1 =
"param1";
    private static final String ARG_PARAM2 =
"param2";

    // TODO: Rename and change types of
parameters
    private String mParam1;
```

```
    private String mParam2;

    public MyFragment() {
        // Required empty public constructor
    }

    /**
     * Use this factory method to create a
new instance of
     * this fragment using the provided
parameters.
     *
     * @param param1 Parameter 1.
     * @param param2 Parameter 2.
     * @return A new instance of fragment
MyFragment.
     */
    // TODO: Rename and change types and
number of parameters
    public static MyFragment
newInstance(String param1, String param2) {
        MyFragment fragment = new
MyFragment();
        Bundle args = new Bundle();
        args.putString(ARG_PARAM1, param1);
        args.putString(ARG_PARAM2, param2);
        fragment.setArguments(args);
        return fragment;
```

```
    }

    @Override
    public void onCreate(Bundle
savedInstanceState) {
        super.onCreate(savedInstanceState);
        if (getArguments() != null) {
            mParam1 =
getArguments().getString(ARG_PARAM1);
            mParam2 =
getArguments().getString(ARG_PARAM2);

        }
        Toast.makeText(getActivity(),
"onCreate", Toast.LENGTH_SHORT).show();
    }

    @Override
    public View onCreateView(LayoutInflater
inflater, ViewGroup container,
                             Bundle
savedInstanceState) {
        Toast.makeText(getActivity(),
"onCreateView()",
Toast.LENGTH_SHORT).show();

        // Inflate the layout for this
fragment
```

```
        return
inflater.inflate(R.layout.fragment_my,
container, false);
    }
    @Override
    public void onViewCreated(@NonNull View
view, @Nullable Bundle savedInstanceState) {
        super.onViewCreated(view,
savedInstanceState);
        Toast.makeText(getActivity(),
"onViewCreated()",
Toast.LENGTH_SHORT).show();
    }
    @Override
    public void onStart() {
        super.onStart();
        Toast.makeText(getActivity(),
"onStart()", Toast.LENGTH_SHORT).show();
    }
    @Override
    public void onResume() {
        super.onResume();
        Toast.makeText(getActivity(),
                "onResume()",
                Toast.LENGTH_SHORT).show();
    }
    @Override
    public void onPause() {
```

```
        super.onPause();
        Toast.makeText(getActivity(),
                "onPause()",
                Toast.LENGTH_SHORT).show();
    }
    @Override
    public void onStop() {
        super.onStop();
        Toast.makeText(getActivity(),
                "onStop()",
                Toast.LENGTH_SHORT).show();
    }
    @Override
    public void onDestroyView() {
        super.onDestroyView();
        Toast.makeText(getActivity(),
"onDestroyView()",
Toast.LENGTH_SHORT).show();
    }
    @Override
    public void onDestroy() {
        super.onDestroy();
        Toast.makeText(getActivity(),
                "onDestroy()",
                Toast.LENGTH_SHORT).show();
    }
    @Override
    public void onDetach() {
```

```
        super.onDetach();
        Toast.makeText(getActivity(),
                "onDetach()",
                Toast.LENGTH_SHORT).show();
    }

}
```

3) activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">

   <fragment
       android:id="@+id/myFragment"
      android:name="com.example.nepnewpract.MyFragment"
       android:layout_width="match_parent"
       android:layout_height="match_parent"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

4) fragment_my.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MyFragment"
   >

   <!-- TODO: Update blank fragment layout -->
   <TextView
       android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:text="Fragment Lifecycle Demo"
        android:textSize="24sp"/>

</LinearLayout>
```

Output:

